

Instructor Guide — Lesson 6: Turn Projects Into Proof

Metadata

- **Path:** Digital Independence Path → Course 16: Build Your Professional Identity → Lesson 6
- **Age range:** 15–18
- **Duration:** 60 minutes
- **Tools used:** Google Docs (case study drafting), GitHub/Vercel (linking live projects), Canva (optional visual polish), HTML/CSS (portfolio embedding)
- **Builds on:** L2 (direction), L3 (portfolio skeleton), L4 (bio + 1 case study), L5 (brand kit)
- **Feeds into:** L7+ (assembling the finished portfolio site)
- **Student deliverable:** 2 additional project case studies (using the L4 template) + a completed Project Selection & Proof Rubric identifying the 2–3 projects that will actually appear on the portfolio

Lesson Overview

Students already have one case study from Lesson 4 and a portfolio skeleton with empty slots. Today they stop treating "projects" and "proof" as the same thing. A project is something they built. Proof is a project explained in a way that lets a stranger — a college admissions reader, a hiring manager, a mentor — understand what problem it solved, what skill it demonstrates, and that it's real. Students will score their available projects against a simple rubric, select their strongest 2–3, and write case studies for two more of them, checking each against the brand kit from Lesson 5 for consistent tone and format.

Prep

- Confirm each student has access to: their L4 case study draft, their L5 brand kit (voice/tone notes, color/font choices), and a running list of everything they've built this course and in personal time (side projects, class assignments, game jams, etc.).
- Print or share digitally: the Project Selection & Proof Rubric worksheet.
- Have the "weak case study → strong rewrite" example ready to display (in the Student Lesson doc) — do not skip reading it aloud; it's the spine of the lesson.
- Optional: pull up 2 real open-source or portfolio project descriptions (one vague, one specific) as a bonus live comparison if time allows.

Teaching Sequence

0:00–0:07 — Hook: "How many projects do you have?" Ask students to shout out a rough count of things they've built. Numbers will range from 2 to 20+. Then ask: "If I only looked at your portfolio for 30 seconds, would I understand what any ONE of these actually does?" Most will admit no. Frame today's lesson: quantity of projects is not the goal — quality of explanation is.

0:07–0:18 — Direct instruction: What makes something "proof"? Cover the three-part test on the board: 1. **Specificity** — names, numbers, real constraints ("I used a stack-based algorithm to solve maze pathfinding in under 200ms" beats "I made a maze game"). 2. **Honesty** — no invented metrics, no claiming a team project as solo work, no "production-ready" for a class assignment. 3. **Demonstrated skill** — the case study should make an invisible decision visible (why you chose a tool, how you debugged something, what tradeoff you made).

Walk through the weak-to-strong rewrite from the Student Lesson as a live example. Have students identify, before you reveal the "strong" version, what's missing from the weak one.

0:18–0:30 — Live demo: scoring a project against the rubric Using a project of your own (real, prepared in advance) or the fictional "Jordan R." example from the Worksheet, model scoring live: - **Complexity (1–5)**: How many moving parts / technical decisions were involved? A static one-page site scores low; a full-stack app with a database scores higher. Complexity is not the same as "hard to build by accident" — reward deliberate technical choices. - **Completeness (1–5)**: Does it work end to end? Is it deployed/shareable? A half-finished idea scores low even if the concept was ambitious. - **What it proves**: One sentence naming the specific skill or trait a viewer would take away. Show how a project can score high on complexity but low on completeness (and therefore might not make the final cut), and vice versa — a simple, fully polished project can beat an ambitious unfinished one.

0:30–0:45 — Guided activity: students score their own projects Students list every candidate project and score it using the Worksheet table. Circulate. Push students who list 6+ projects to be honest about completeness scores — this is where "more is better" thinking surfaces. Push students who list only 1–2 projects to check personal projects outside of Lumexa class time (a phone app, a mod, a website for a family business, a school project).

0:45–0:55 — Independent work: draft 2 more case studies Using the L4 template (Project Name, Problem/Goal, Tools Used, My Approach, Result, What I Learned), students draft case studies for their 2nd and 3rd highest-scoring projects. Remind them to cross-check tone/vocabulary against their L5 brand kit — the case studies should sound like the same person wrote all three.

0:55–1:00 — Lesson close Cold-call 2–3 students to share one line from a new case study. Reinforce: by next lesson they should have 2–3 finished case studies and know exactly which projects are "in" and which are "out." Preview L7: assembling these into the actual portfolio site.

Discussion Questions

1. Think of a project you're proud of that would actually score LOW on this rubric. What would it take to raise its completeness score?
2. Why might a simple project with a great explanation beat a complex project with no explanation?

3. What's the difference between "making myself sound impressive" and "being honest about what I did"? Can both be true at once?
4. If a stranger read only your case study — not your code — would they understand what you actually did versus what a tutorial or teammate did?

Common Misconceptions

- **"More projects on my portfolio = more impressive."** Reality: reviewers spend seconds per project. Three well-explained projects beat eight one-line ones. A crowded portfolio actually buries the good work.
- **"I need something impressive, not something honest."** Reality: vague inflated claims are a red flag to experienced readers (teachers, recruiters) and fall apart under a single follow-up question ("tell me more about that"). Specific + honest is more persuasive than impressive + vague, because it's checkable and it shows real thinking.
- **"If I didn't finish it, I shouldn't score it low, since the idea was good."** Reality: an idea is not proof. Completeness matters because proof requires something a viewer can actually see or use.

Student Difficulties & Troubleshooting

- **"I don't have 3 good projects."** Solution: broaden the search — school assignments, personal projects, group projects (with honest attribution of their specific role) all count. Two strong case studies beat three where one is weak filler.
- **Students inflate scores to justify keeping a favorite project.** Solution: have a peer (or you) sanity-check one score per student out loud — ask "walk me through why this is a 4, not a 2."
- **Case studies read identically to the Lesson 4 one in structure but not tone.** Solution: point back to the brand kit's specific word choices/voice notes, not just colors/fonts.
- **Students conflate "complex" with "long."** Redirect to: complexity is about decisions made, not lines of code or word count.

How to Review

Check each student's Worksheet for: at least 4 candidate projects scored, honest-looking scores (not all 5s), and a clear "include/exclude" decision with reasoning that matches the scores. Check both new case studies against the L4 template fields (all six must be filled) and against the three-part proof test (specific, honest, demonstrates a skill). Flag any case study using vague language ("worked on," "helped with," "involved") without a concrete detail nearby.

Lesson Close

Students leave with 2 new case studies and a rubric-backed decision on their final 2–3 featured projects, ready to slot into the portfolio structure in the next lesson.